For Examiner's Use

8 **(a)** Fig. 8.1 shows an α-particle A as it approaches and passes by a stationary gold nucleus N.

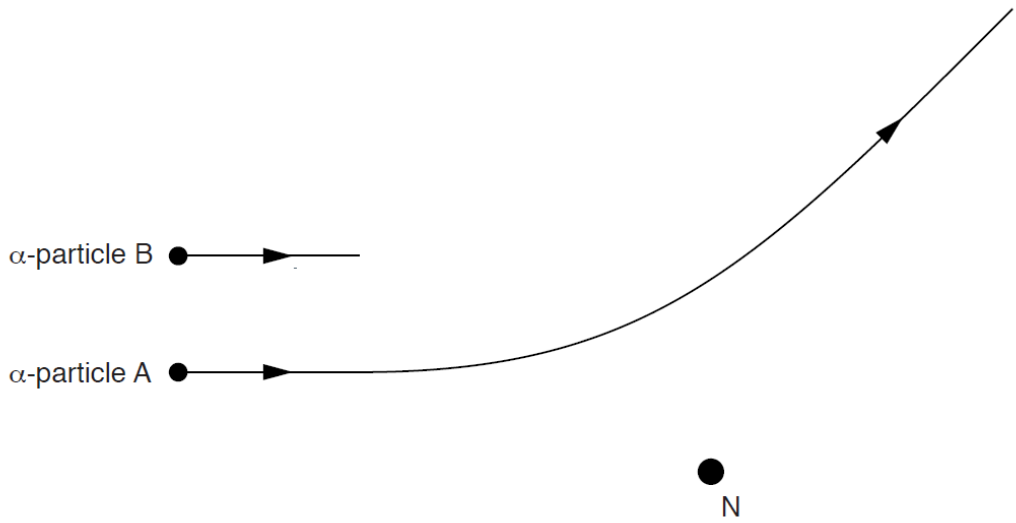

Fig. 8.1

A second α-particle B has the same initial direction and energy as α-particle A.

On Fig. 8.1, complete the path of α-particle B as it approaches and passes by the nucleus N. [2]

(b) An alpha particle has a speed of 1.30×10^7 m s^{-1}.

(i) Calculate the kinetic energy of the alpha particle.

kinetic energy = ... J [2]

(ii) The alpha particle is aimed directly at a gold nucleus which has a proton number of 79.

Calculate the distance of closest approach *r*.

r = ... m [3]

For Examiner's Use

(c) A radiation detector is placed close to a radioactive source. The detector does not surround the source.

Radiation is emitted in all directions and, as a result, the activity of the source and the measured count rate are different.

Suggest two other reasons why the activity and the measured count rate may be different.

1. .....

.....

2. .....

..... [2]

(d) The variation with time *t* of the measured count rate in **(c)** is shown in Fig. 8.2.

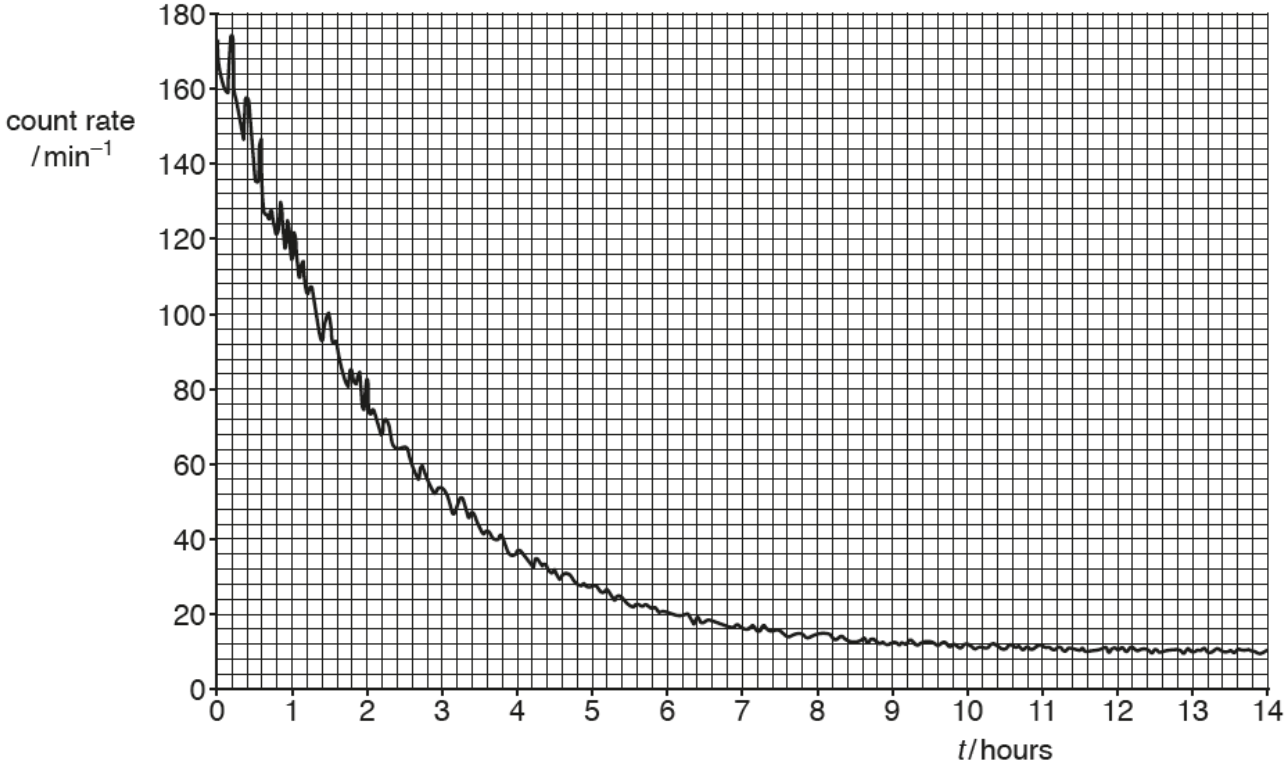

Fig. 8.2

(i) State the feature of Fig. 8.2 that indicates the random nature of radioactive decay.

.....

..... [1]

For Examiner's Use

(ii) Use Fig. 8.2 to determine the half-life of the radioactive isotope in the source.

half-life = ... hours [4]

(e) The readings in **(d)** were obtained at room temperature. A second sample of this isotope is heated to a temperature of 500 °C. The initial count rate at time t = 0 is the same as that in **(d)**. The variation with time t of the measured count rate from the heated source is determined.

State, with a reason, the difference, if any, in

1. the half-life,

……...

……...

……...

2. the measured count rate for any specific time.

……...

……...

…….. [3]

For Examiner's Use

(f) A small volume of solution containing the radioactive isotope sodium-24 ($^{24}_{11}Na$) has an initial activity of 3.8×10^4 Bq. Sodium-24, of half-life 15 hours, decays to form a stable daughter isotope.

All of the solution is poured into a container of water. After 36 hours, a sample of water of volume 5.0 cm^3, taken from the container, is found to have an activity of 1.2 Bq.

Assuming that the solution of the radioactive isotope is distributed uniformly throughout the container of water, calculate the volume of water in the container.

volume = ... cm^3 [3]